

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

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Data Folder: data/MATH500/deepseek/p1
Total Items: 500
Total Strategies: 9

OVERALL ACCURACY

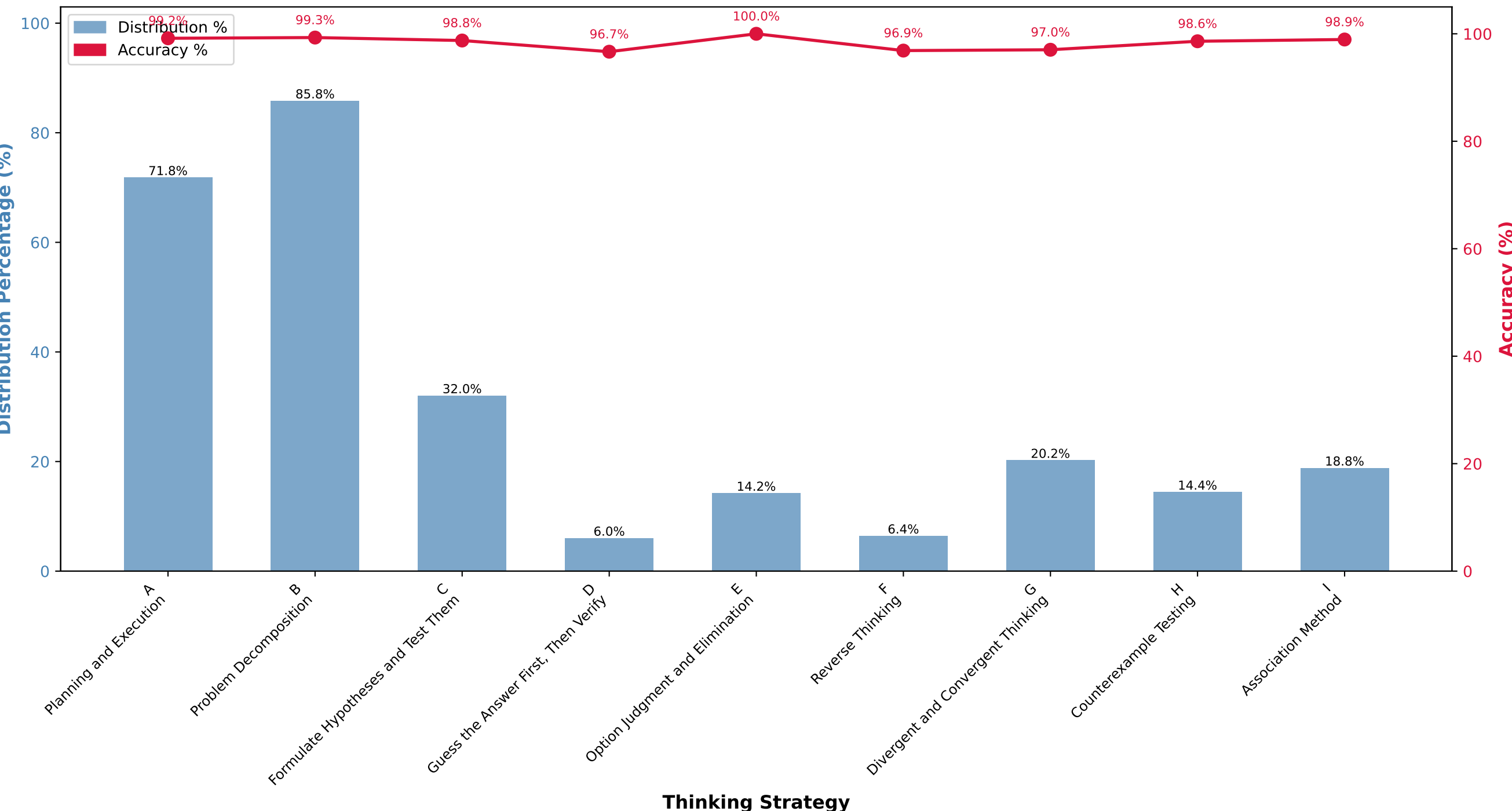
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Total Problems: 500
Correct: 496
Incorrect: 4
Overall Accuracy: 99.20%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 71.8%
Accuracy: 99.16% (356/359)
 - B: Problem Decomposition
Distribution: 85.8%
Accuracy: 99.30% (426/429)
 - C: Formulate Hypotheses and Test Them
Distribution: 32.0%
Accuracy: 98.75% (158/160)
 - D: Guess the Answer First, Then Verify
Distribution: 6.0%
Accuracy: 96.67% (29/30)
 - E: Option Judgment and Elimination
Distribution: 14.2%
Accuracy: 100.00% (71/71)
 - F: Reverse Thinking
Distribution: 6.4%
Accuracy: 96.88% (31/32)
 - G: Divergent and Convergent Thinking
Distribution: 20.2%
Accuracy: 97.03% (98/101)
 - H: Counterexample Testing
Distribution: 14.4%
Accuracy: 98.61% (71/72)
 - I: Association Method
Distribution: 18.8%
Accuracy: 98.94% (93/94)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

=====

Write $\frac{3}{20}$ as a decimal.

GROUND TRUTH ANSWER

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0.15

MODEL RESPONSE

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To convert $\frac{3}{20}$ to a decimal, divide 3 by 20:

$$\frac{3}{20} = 3 \div 20 = 0.15$$

Alternatively, note that $\frac{3}{20} = \frac{15}{100}$, which is equivalent to 0.15.

Thus, $\frac{3}{20} = 0.15$.

CHAIN OF THOUGHT (First 1500 chars)

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Okay, the user asked to convert the fraction $\frac{3}{20}$ into a decimal. Let me start by breaking down how to do this.

First, I need to understand what the fraction means. $\frac{3}{20}$ is the same as 3 divided by 20. So, I'll perform that division.

Since 20 is larger than 3, I'll add a decimal point and zeros to 3, making it 3.0, then 3.00, and so on. Dividing 3.00 by 20 gives me 0.15 because 20 times 0.15 is exactly 3.00.

I should also consider if there's a simpler way. I know that 20 times 5 is 100, so multiplying both numerator and denominator by 5 gives $\frac{15}{100}$, which is 0.15 directly. That's a quicker method.

Double-checking: 0.15 times 20 is indeed 3. So, the decimal is correct. No need for rounding since it terminates neatly.

Finally, present the answer clearly as 0.15.

Example for Strategy B: Problem Decomposition

PROBLEM

Compute $\sin^3 18^\circ + \sin^2 18^\circ$.

GROUND TRUTH ANSWER

$\frac{1}{8}$

MODEL RESPONSE

Step 1. Find $\sin 18^\circ$

It is known that:

$$\sin 18^\circ = \frac{\sqrt{5} - 1}{4}$$

Step 2. Compute $\sin^2 18^\circ$

$$\sin^2 18^\circ = \left(\frac{\sqrt{5} - 1}{4} \right)^2 = \frac{(\sqrt{5} - 1)^2}{16} = \frac{5 - 2\sqrt{5} + 1}{16} = \frac{6 - 2\sqrt{5}}{16} = \frac{3 - \sqrt{5}}{8}$$

Step 3. Compute $\sin^3 18^\circ$

$$\sin^3 18^\circ = \sin 18^\circ \cdot \sin^2 18^\circ = \frac{\sqrt{5} - 1}{4} \cdot \frac{3 - \sqrt{5}}{8} = \frac{(\sqrt{5} - 1)(3 - \sqrt{5})}{32}$$

Expand the numerator:

$$(\sqrt{5} - 1)(3 - \sqrt{5}) = 3\sqrt{5} - 5 - 3 + \sqrt{5} = 4\sqrt{5} - 8$$

So,

$$\sin^3 18^\circ = \frac{4\sqrt{5} - 8}{32} = \frac{\sqrt{5} - 2}{8}$$

Step 4. Add...

CHAIN OF THOUGHT (First 1500 chars)

We are asked: "Compute $\sin^3 18^\circ + \sin^2 18^\circ$." Possibly it's $\sin^3 18^\circ + \sin^2 18^\circ$? But careful: "Compute $\sin^3 18^\circ + \sin^2 18^\circ$." Possibly it's $\sin^3 18^\circ + \sin^2 18^\circ$? But then it's ambiguous: It might be $\sin^3(18^\circ) + \sin^2(18^\circ)$? But then it's simply $\sin^2(18^\circ)(\sin(18^\circ)+1)$? But then maybe it simplifies to something? Alternatively, it might be $\sin^3(18^\circ) + \sin^2(18^\circ)$? Possibly it's a misprint? Alternatively, it might be $\sin^3 18^\circ + \sin^2 18^\circ$? Possibly it's $\sin^3(18^\circ) + \sin^2(18^\circ)$? But then we know $\sin(18^\circ) = \frac{\sqrt{5}-1}{4}$? Actually $\sin(18^\circ) = \frac{\sqrt{5}-1}{4}$? Wait: Actually $\sin(18^\circ) = \frac{\sqrt{5}-1}{4}$? Check: $\sin(18^\circ) = 0.309016\dots$ and $\frac{\sqrt{5}-1}{4} = \frac{2.23607-1}{4} = 1.23607/4 = 0.3090175$. So yes.

Then $\sin^2(18^\circ) = \left(\frac{\sqrt{5}-1}{4} \right)^2 = \frac{6-2\sqrt{5}}{16} = \frac{3-\sqrt{5}}{8}$.
And $\sin^3(18^\circ) = \sin(18^\circ) \cdot \sin^2(18^\circ) = \frac{\sqrt{5}-1}{4} \cdot \frac{3-\sqrt{5}}{8} = \frac{(\sqrt{5}-1)(3-\sqrt{5})}{32} = \frac{3\sqrt{5} - 5 - 3 + \sqrt{5}}{32} = \frac{4\sqrt{5} - 8}{32} = \frac{\sqrt{5} - 2}{8}$.
Then sum: $\sin^3(18^\circ) + \sin^2(18^\circ) = \frac{\sqrt{5}-2}{8} + \frac{3-\sqrt{5}}{8} = \frac{\sqrt{5}-2+3-\sqrt{5}}{8} = \frac{1}{8}$.

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

=====

\$n\$ fair 6-sided dice are simultaneously rolled. The probability that exactly two of them show a number other than 1 is $\frac{25}{216}$ \$. Find \$n\$.

GROUND TRUTH ANSWER

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4

MODEL RESPONSE

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Step 1: Define the Event

We roll n fair 6-sided dice. The event of interest is that **exactly two** dice show a number **other than 1** (i.e., a number from $\{2, 3, 4, 5, 6\}$), and the remaining $n - 2$ dice show a **1**.

Step 2: Compute the Probability

- Probability that a die shows a number other than 1:
 $\frac{5}{6}$
- Probability that a die shows a 1:
 $\frac{1}{6}$

The number of ways to choose which 2 dice show a number other than 1 is given by the binomial coefficient:

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

So, the probability is:

$$P = \binom{n}{2} \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^{n-2}$$

Step 3: Set Up the Equation

Given:

$$\frac{n(n-1)}{2} \cdot \frac{25}{36} = \frac{25}{216}$$

CHAIN OF THOUGHT (First 1500 chars)

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We are given: "\$n\$ fair 6-sided dice are simultaneously rolled. The probability that exactly two of them show a number other than 1 is $\frac{25}{216}$ \$. Find \$n\$."

Interpretation: "exactly two of them show a number other than 1" means: Among n dice, exactly two dice show numbers from $\{2,3,4,5,6\}$ (i.e., not 1) and the remaining $(n-2)$ dice show 1. But careful: "exactly two of them show a number other than 1" might also mean that exactly two dice show numbers that are not 1. But then the probability is: $C(n,2) * (5/6)^2 * (1/6)^{(n-2)}$? But then that equals given: $(25/216)$. So we have: $C(n,2) * (5/6)^2 * (1/6)^{(n-2)} = 25/216$.
That is: $(n(n-1)/2) * (25/36) * (1/6)^{(n-2)} = 25/216$.
Simplify: $(n(n-1)/2)*25/36 = (25 n(n-1))/(72)$. Then multiplied by $(1/6)^{(n-2)}$ gives: $25 n(n-1)/(72) * (1/6)^{(n-2)} = 25/216$.
Cancel 25 (assuming nonzero): $n(n-1)/(72) * (1/6)^{(n-2)} = 1/216$.
Multiply both sides by 72: $n(n-1) * (1/6)^{(n-2)} = 72/216 = 1/3$.
So equation: $n(n-1) / 6^{(n-2)} = 1/3$. Multiply both sides by 3: $3 n(n-1) = 6^{(n-2)}$. So equation: $6^{(n-2)} = 3 n(n-1)$.

Now test small n :

$n=2$: LHS: $6^0=1$, RHS: $3*2*1=6$, no.
 $n=3$: LHS: $6^1=6$, RHS: $3*3*2=18$, no.
 $n=4$: LHS: $6^2=36$, RHS: $3*4*3=36$, so $n=4$ works.
 $n=5$: LHS: $6^3=216$, RHS: $3*5*4=60$, no.
 $n=6$: LHS: $6^4=1296$, RHS: $3*6*5=90$, no.
So answer: $n=4$.

Thus answer: $\boxed{4}$.

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

=====

\$n\$ fair 6-sided dice are simultaneously rolled. The probability that exactly two of them show a number other than 1 is $\frac{25}{216}$ \$. Find \$n\$.

GROUND TRUTH ANSWER

=====

4

MODEL RESPONSE

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Step 1: Define the Event

We roll n fair 6-sided dice. The event of interest is that **exactly two** dice show a number **other than 1** (i.e., a number from $\{2, 3, 4, 5, 6\}$), and the remaining $n - 2$ dice show a **1**.

Step 2: Compute the Probability

- Probability that a die shows a number other than 1:
 $\frac{5}{6}$
- Probability that a die shows a 1:
 $\frac{1}{6}$

The number of ways to choose which 2 dice show a number other than 1 is given by the binomial coefficient:

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

So, the probability is:

$$P = \binom{n}{2} \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^{n-2}$$

Step 3: Set Up the Equation

Given:

$$\frac{n(n-1)}{2} \cdot \frac{25}{36} = \frac{25}{216}$$

CHAIN OF THOUGHT (First 1500 chars)

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We are given: "\$n\$ fair 6-sided dice are simultaneously rolled. The probability that exactly two of them show a number other than 1 is $\frac{25}{216}$ \$. Find \$n\$."

Interpretation: "exactly two of them show a number other than 1" means: Among n dice, exactly two dice show numbers from $\{2,3,4,5,6\}$ (i.e., not 1) and the remaining $(n-2)$ dice show 1. But careful: "exactly two of them show a number other than 1" might also mean that exactly two dice show numbers that are not 1. But then the probability is: $C(n,2) * (5/6)^2 * (1/6)^{(n-2)}$? But then that equals given: $(25/216)$. So we have: $C(n,2) * (5/6)^2 * (1/6)^{(n-2)} = 25/216$. That is: $(n(n-1)/2) * (25/36) * (1/6)^{(n-2)} = 25/216$. Simplify: $(n(n-1)/2)*25/36 = (25 n(n-1))/(72)$. Then multiplied by $(1/6)^{(n-2)}$ gives: $25 n(n-1)/(72) * (1/6)^{(n-2)} = 25/216$. Cancel 25 (assuming nonzero): $n(n-1)/(72) * (1/6)^{(n-2)} = 1/216$. Multiply both sides by 72: $n(n-1) * (1/6)^{(n-2)} = 72/216 = 1/3$. So equation: $n(n-1) / 6^{(n-2)} = 1/3$. Multiply both sides by 3: $3 n(n-1) = 6^{(n-2)}$. So equation: $6^{(n-2)} = 3 n(n-1)$.

Now test small n :

$n=2$: LHS: $6^0=1$, RHS: $3*2*1=6$, no.
 $n=3$: LHS: $6^1=6$, RHS: $3*3*2=18$, no.
 $n=4$: LHS: $6^2=36$, RHS: $3*4*3=36$, so $n=4$ works.
 $n=5$: LHS: $6^3=216$, RHS: $3*5*4=60$, no.
 $n=6$: LHS: $6^4=1296$, RHS: $3*6*5=90$, no.
So answer: $n=4$.

Thus answer: $\boxed{4}$.

Example for Strategy E: Option Judgment and Elimination

PROBLEM

A strictly increasing sequence of positive integers $a_1, a_2, a_3, \dots$ has the property that for every positive integer k , the subsequence $a_{2k-1}, a_{2k}, a_{2k+1}$ is geometric and the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is arithmetic. Suppose that $a_{13} = 2016$. Find a_1 .

GROUND TRUTH ANSWER

504

MODEL RESPONSE

Step 1. Understand the Conditions

The sequence $(a_1, a_2, a_3, \dots)$ is strictly increasing and consists of positive integers. For every positive integer k :

- The triple $(a_{2k-1}, a_{2k}, a_{2k+1})$ forms a geometric progression.
- The triple $(a_{2k}, a_{2k+1}, a_{2k+2})$ forms an arithmetic progression.

Also, $a_{13} = 2016$. Our goal is to find a_1 .

Step 2. Express the Terms in Terms of a_1 and a Ratio

Let:

$$r = \frac{a_2}{a_1}$$

Since (a_1, a_2, a_3) form a geometric progression:

$$a_2 = r a_1, \quad a_3 = r^2 a_1$$

Now, (a_2, a_3, a_4) form an arithmetic progression:

$$2a_3 = a_2 + a_4 \implies a_4 = 2a_3 - a_2 = 2r^2 a_1 - r a_1 = a_1 r (2r - 1)$$

Next, (a_3, a_4, a_5)

CHAIN OF THOUGHT (First 1500 chars)

We are given: "A strictly increasing sequence of positive integers $a_1, a_2, a_3, \dots$ has the property that for every positive integer k , the subsequence $a_{2k-1}, a_{2k}, a_{2k+1}$ is geometric and the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is arithmetic. Suppose that $a_{13} = 2016$. Find a_1 ."

Let's denote terms: $a_1, a_2, a_3, a_4, a_5, a_6, \dots$

Properties: For every positive integer k :

(1) $a_{2k-1}, a_{2k}, a_{2k+1}$ is geometric. So: $a_{2k}^2 = a_{2k-1} \cdot a_{2k+1}$? Actually careful: Geometric means there is a common ratio r such that $a_{2k} = r \cdot a_{2k-1}$ and $a_{2k+1} = r^2 \cdot a_{2k-1}$? But careful: It says "the subsequence is geometric", so it means that $a_{2k}/a_{2k-1} = a_{2k+1}/a_{2k}$? Actually, if three numbers are in geometric progression, then the middle is the geometric mean: $a_{2k}^2 = a_{2k-1} \cdot a_{2k+1}$. But it doesn't necessarily mean that the ratio is constant? Actually, if a, b, c are in geometric progression, then $b/a = c/b$. So indeed, $a_{2k}/a_{2k-1} = a_{2k+1}/a_{2k}$. So let that common ratio be r_k . But careful: It might be different for different k . So we denote: For k , let $r_k = a_{2k}/a_{2k-1} = a_{2k+1}/a_{2k}$. So then $a_{2k} = r_k a_{2k-1}$ and $a_{2k+1} = r_k^2 a_{2k-1}$.

(2) For every positive integer k , the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is arithmetic. So that means: $2a_{2k+1} = a_{2k} + a_{2k+2}$ or equivalently $a_{2k+2} = 2a_{2k+1} - a_{2k}$. Alternatively, the common difference d_k such that $a_{2k+1} = a_{2k} + d_k$ and $a_{2k+2} = a_{2k+1} + d_k$.

Example for Strategy F: Reverse Thinking

PROBLEM

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In the diagram, two circles, each with center D , have radii of 1 and 2 . The total area of the shaded region is $\frac{5}{12}$ of the area of the larger circle. How many degrees are in the measure of (the smaller) $\angle ADC$?

[asy
size(4cm);
defaultpen(linewidth(0.75));
real adc = 100;
pair d = (0, 0); pair a = 2 * dir(100); pair c = (2, 0);
path inner = arc(d, a/2, c/2, CW);

path outer = arc(d, c, a, CCW);
guide region1 = (a--a/2)..inner..(c/2--c)..outer..cycle;
guide region2...

GROUND TRUTH ANSWER

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120

MODEL RESPONSE

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□ Step 1: Understand the Diagram

We have two concentric circles centered at point D , with radii 1 and 2 . Points A and C lie on the larger circle such that $\angle ADC = \theta$. The shaded region consists of two parts:

- **Region 1**: The annular sector between the two circles, bounded by the two radii and the arcs on both circles.
- **Region 2**: The part of the smaller circle **not** included in the sector of angle θ .

□ Step 2: Compute the Shaded Area

(a) Area of Region 1

This is the area of the annular sector between the two circles:

\\[
\\text{Area}_1 = \\frac{\\theta}{360^\\circ} \\cdot (\\pi \\cdot 2^2 - \\pi \\cdot 1^2) = \\frac{\\theta}{360^\\circ} \\cdot 3\\pi
\\]

(b) Area of Region 2

This is the ar...

CHAIN OF THOUGHT (First 1500 chars)

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We are given: "In the diagram, two circles, each with center D , have radii of 1 and 2 . The total area of the shaded region is $\frac{5}{12}$ of the area of the larger circle. How many degrees are in the measure of (the smaller) $\angle ADC$?"

Interpretation: There is diagram: two circles centered at D . One circle radius 1 , one circle radius 2 . There is points A and C . Possibly A on outer circle? Actually: "two circles, each with center D , have radii of 1 and 2 ." So inner circle radius 1 , outer circle radius 2 . Then: "The total area of the shaded region is $\frac{5}{12}$ of the area of the larger circle." That means: Shaded area = $(\frac{5}{12}) \cdot \text{Area}(\text{larger circle}) = (\frac{5}{12}) \cdot \pi \cdot 2^2 = (\frac{5}{12}) \cdot 4\pi = (\frac{5\pi}{3})$? But careful: Actually: $(\frac{5}{12}) \cdot \text{area of larger circle} = (\frac{5}{12}) \cdot 4\pi = (\frac{5\pi}{3})$. But then we want: "How many degrees are in the measure of (the smaller) $\angle ADC$?" So angle ADC is at center D . Points A and C are on the circles? Possibly: A on outer circle? Actually from diagram: labels: A, D, C . D is center. A and C are on the boundary? Possibly: A is on the outer circle? But then there is also inner circle? The diagram: "two circles, each with center D , have radii of 1 and 2 ." So there is a small circle and a big circle. Then there is angle ADC . That angle likely cuts through both circles. And there is shading. The shading: There are two shaded regions. One is the region that is inside the larger circle but outside the smaller circle? But then: "The total area of the shaded region..."

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

The polynomial $x^3 - 3x^2 + 4x - 1$ is a factor of $x^9 + px^6 + qx^3 + r$. Enter the ordered triple (p,q,r) .

GROUND TRUTH ANSWER

(6,31,-1)

MODEL RESPONSE

Step 1. Understand the Problem

We are given that the polynomial

$$g(x) = x^3 - 3x^2 + 4x - 1$$

is a factor of

$$f(x) = x^9 + p x^6 + q x^3 + r.$$

This means that for every root α of $g(x)$, we must have $f(\alpha) = 0$.

Step 2. Express $f(\alpha)$ in Terms of Lower Powers

Let α be a root of $g(x)$, so:

$$\alpha^3 - 3\alpha^2 + 4\alpha - 1 = 0 \quad \Rightarrow \quad \alpha^3 = 3\alpha^2 - 4\alpha + 1.$$

We want to compute:

$$f(\alpha) = \alpha^9 + p \alpha^6 + q \alpha^3 + r.$$

Using equation (1), we can express higher powers of α in terms of lower powers.

Step 3. Compute $\alpha^4, \alpha^6, \alpha^9$

- **Compute α^4 :**
...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "The polynomial $x^3 - 3x^2 + 4x - 1$ is a factor of $x^9 + px^6 + qx^3 + r$. Enter the ordered triple (p,q,r) ."

Interpretation: There exists polynomial $f(x) = x^9 + p x^6 + q x^3 + r$. And it is divisible by $g(x) = x^3 - 3x^2 + 4x - 1$. So $f(x) \bmod g(x) = 0$.

We want p,q,r .

Observing: $f(x) = (x^3)^3 + p (x^3)^2 + q (x^3) + r$. So it's a polynomial in x^3 . Alternatively, we can set $y = x^3$. Then $f(x) = y^3 + p y^2 + q y + r$. But careful: That representation is not exactly a polynomial in y because x appears in $g(x)$? Actually, $g(x)$ is cubic. But maybe there is relation: If $g(x)$ divides $f(x)$, then for every root α of $g(x)$, $f(\alpha) = 0$. So if α is a root of $g(x) = x^3 - 3x^2 + 4x - 1 = 0$, then $\alpha^3 - 3\alpha^2 + 4\alpha - 1 = 0$. But note that $\alpha^3 = 3\alpha^2 - 4\alpha + 1$. Then maybe we can compute higher powers mod $g(x)$. Alternatively, we can compute the polynomial that has roots as cubes of the roots of $g(x)$? Because $f(x)$ is in terms of x^3 . So if α is a root of g , then $\beta = \alpha^3$. Then $f(\alpha) = \beta^3 + p \beta^2 + q \beta + r = 0$. So if we let the three roots of g be $\alpha_1, \alpha_2, \alpha_3$, then their cubes β_i satisfy a polynomial: $\beta^3 + p \beta^2 + q \beta + r = 0$. And we want p,q,r . And they are symmetric functions of β_i . But then we need to compute from $g(x)=0$. Alternatively, we can compute the minimal polynomial of β from α . But careful: α satisfies $\alpha^3 = 3\alpha^2 - 4\alpha + 1$. So then $\beta = \alpha^3$. Then we can compute $\alpha^6 = (\alpha^3)^2 = (3\alpha^2 - 4\alpha + 1)^2 = 9\alpha^4 - 24\alpha^3 + 24\alpha^2 - 8\alpha + 1$.

Example for Strategy H: Counterexample Testing

PROBLEM

Find the number of quadratic equations of the form $x^2 + ax + b = 0$, such that whenever c is a root of the equation, $c^2 - 2$ is also a root of the equation.

GROUND TRUTH ANSWER

6

MODEL RESPONSE

Step 1. Understand the Condition

We consider quadratic equations of the form:

$$x^2 + ax + b = 0$$

Let the roots be r and s (possibly equal). The condition is:

> Whenever c is a root, $c^2 - 2$ is also a root.

Define the function:

$$f(x) = x^2 - 2$$

Then the condition becomes:

$$f(r) \in \{r, s\} \quad \text{and} \quad f(s) \in \{r, s\}$$

Step 2. Analyze the Possibilities

There are three main cases:

Case 1: Both Roots Are Fixed Points

If $f(r) = r$ and $f(s) = s$, then:

$$\begin{aligned} r^2 - 2 &= r \implies r^2 - r - 2 = 0 \implies r = 2 \text{ or } -1 \\ s^2 - 2 &= s \implies s^2 - s - 2 = 0 \implies s = 2 \text{ or } -1 \end{aligned}$$

So the possible pairs (r, s) are:

- $(2, 2)$

CHAIN OF THOUGHT (First 1500 chars)

We are asked: "Find the number of quadratic equations of the form $x^2 + ax + b = 0$, such that whenever c is a root of the equation, $c^2 - 2$ is also a root of the equation."

Interpretation: We have quadratic equations: $x^2 + ax + b = 0$. They have two roots (maybe repeated, but then condition: "whenever c is a root, $c^2 - 2$ is also a root." That means: For each root c , the number $f(c) = c^2 - 2$ is also a root. Possibly the roots could be the same? Possibly the condition applies to both roots. So if r and s are roots, then we require: $f(r)$ and $f(s)$ are roots. But careful: "such that whenever c is a root, $c^2 - 2$ is also a root." That means: For every root c , $f(c)$ is in the set of roots. So if the roots are r and s , then $f(r)$ and $f(s)$ are either r or s . So we have conditions: $f(r) \in \{r, s\}$ and $f(s) \in \{r, s\}$.

Also note: The quadratic is of the form $x^2 + ax + b = 0$. So a and b are real? Possibly? The problem statement doesn't specify real coefficients? Usually quadratic equations are with real coefficients? But it says "quadratic equations" so likely a and b are real numbers. But then the roots could be complex. But then condition " $c^2 - 2$ is also a root. Possibly we consider equations with real coefficients? The problem statement: "Find the number of quadratic equations of the form $x^2 + ax + b = 0$ " It doesn't say "with real coefficients" explicitly. But often it's implied that a and b are real. But then the condition "whenever c is a root, $c^2 - 2$ is also a root" might force...

Example for Strategy I: Association Method

PROBLEM

```
=====
In circle $J$, $HO$ and $HN$ are tangent to the circle at $O$ and $N$. Find the number of degrees in the sum of  $\angle J$  and  $\angle H$ . [asy]
size(150);
import graph;
pair J = (0,0), H = (6,0), O, N;
path circ = Circle(J,3);
pair M = midpoint(J--H);
path secCirc = Circle(M,3);
pair[] tangentPoints = intersectionpoints(circ,secCirc);
O = tangentPoints[0]; N = tangentPoints[1];
draw(J--N--H--O--cycle);
draw(circ);
label("$H$",H,E);
label("$J$",J,W);
label("$N$",N,S);
lab...
```

GROUND TRUTH ANSWER

180 $^\circ$

MODEL RESPONSE

```
### Step 1: Understand the Diagram

- Circle with center  $J$ .
- Point  $H$  lies outside the circle.
- From  $H$ , two tangents are drawn to the circle, touching it at points  $O$  and  $N$ .
- The quadrilateral  $H O J N$  is formed by connecting points  $H, O, J, N$ .
```

Step 2: Identify the Angles

We are asked to find the sum:

```
[[
 $\angle J + \angle H$ 
]]
```

where:

- $\angle J$ is the angle at vertex J (between segments JO and JN)
- $\angle H$ is the angle at vertex H (between segments HO and HN)

Step 3: Use Tangent–Radius Properties

A key property of circles:

> A tangent to a circle is perpendicular to the radius drawn to the point of tangency.

Thus:

- ...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "In circle J , HO and HN are tangent to the circle at O and N . Find the number of degrees in the sum of $\angle J$ and $\angle H$."

Interpretation: There is circle with center J . Points O and N on circle. Lines from H to O and H to N are tangents. So triangle HON ? Actually: " HO and HN are tangent to the circle at O and N ." So H is external point. Then we have triangle HON ? But then there is angle J and angle H . Possibly these are angles at vertices J and H in quadrilateral $HOJN$? Actually: The diagram: [asy] code:

```
size(150);
import graph;
pair J = (0,0), H = (6,0), O, N;
path circ = Circle(J,3);
pair M = midpoint(J--H);
path secCirc = Circle(M,3);
pair[] tangentPoints = intersectionpoints(circ,secCirc);
O = tangentPoints[0]; N = tangentPoints[1];
draw(J--N--H--O--cycle);
draw(circ);
label("$H$",H,E);
label("$J$",J,W);
label("$N$",N,S);
label("$O$",O,NE);
[/asy]
```

So: Circle centered at J . H is a point outside. From H , two tangents to circle: one touches at O , the other at N . So lines HO and HN are tangents. Then we draw lines: $J--N$, $H--O$, etc. Actually: The drawing: `draw(J--N--H--O--cycle)`. So quadrilateral: J, N, H, O . So we have two triangles: JNO and JHO ? Actually, it's a quadrilateral with vertices J, N, H, O . And we are asked: "Find the number of degrees in the sum of $\angle J$ and $\angle H$." That is sum of angles at vertex J and vertex H in this quadrilateral.

Let's denote: $\angle J$ is at center J ? Actually careful: " $\angle J$ " likely means ang...